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24DS742 - Reinforcement Learning
Ph.D.
Course Plan and Evaluation Scheme

Course Objectives
Learn how to define RL tasks and the core principals behind the RL
Understand and work with approximate solutions (deep Q network-based algorithms)
Explore imitation learning tasks and solution
Recognize current advanced techniques and applications in RL

Course Outcomes
CO1: Understand the relevance of Reinforcement Learning and how does it complement other ML techniques
CO2: Understand various RL algorithms
CO3: Formulate a problem as a RL problem and solve it
CO4: Implement RL algorithms

CO-PO Mapping

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PSO1	PSO2	PSO3
CO1	3	1	2	1	-	-	-	-	2
CO2	3	1	2	1	-	-	3	-	2
CO3	-	3	2	3	3	-	-	-	3
CO4	3	3	3	3	-	-	-	3	3

Syllabus
Reinforcement learning vs all, multi-armed bandit, Decision process & applications, Markov Decision Process, Cross entropy method, Approximate cross entropy method, approximate cross entropy method, Evolution strategies: core idea, math problems, log-derivative trick, duct tape. Blackbox optimization: drawback - Dynamic Programming, Reward design, state and Action Value Functions, Measuring Policy Optimality, Policy: evaluation & improvement, Policy and value iteration, Model-free methods: Model-based vs model-free, Monte-Carlo & Temporal Difference; Q-learning, Exploration vs Exploitation, Footnote: Monte-Carlo vs Temporal Difference, Accounting for exploration. Expected Value SARSA, On-policy vs off-policy, Experience replay. Approximate Value Based Methods: Supervised & Reinforcement Learning, Loss functions in value-based RL, difficulties with Approximate Methods, DQN – bird's eye view, DQN – the internals, DQN: statistical issues, Double Q-learning, More DQN tricks, Partial observability. RLHF- Reinforcement Learning through Human feedback

Textbooks / References

• *Sutton and Barto, Reinforcement Learning: An Introduction, 2nd Edition. MIT Press, Cambridge, MA, 2018*
• *Csaba Szepesvári, Algorithms for Reinforcement Learning, Morgan & Claypool. 2010.*
• *Marco Wiering and Martijn van Otterlo, Reinforcement Learning: State-of-the-Art*
•*Adaptation, Learning, and Optimization, Springer, 2012.*

Evaluation Pattern

Assessment component	Type of Assessment	Minimum Number of Assessments	Weightage (%)
Internal	Quizzes	2	10
Internal	Assignments/Presentations	3	20
Internal	Mid-Term Exam	1	20
External	End Semester Exam(Term Project)	1	30
External	End Semester Exam	1	20

Lecture Plan

Week	Lecture No(s)	Topics	Keywords	Theory(T)/Lab(L)	Course Outcomes	Resource
1	1-2	Introduction to RL	Introduction to course, RL Defined, Comparison with other Machine Learning techniques and RL examples	T	CO1	B1 and Online
1	3	Introduction to RL	What is RL? Why RL? Key Feature of RL, History, Realtime Analogy to RL	T	CO1	B1 and Online
2	4-5	Elements of RL	Elements of RL and Tic-Tac-Toe Example	T	CO1	B1
2	6-7	Tic-Tac-Toe and Open AI Gym Exploration	Understanding Tic-Tac-Toe RL agent implementation and Exploring various Open AI Gym Environments	L	CO2	Online
3	8	RL Terminologies and RL formulation Examples	Terminologies: Time Step, State, Action, Reward, Observation, Model, and Policy, Exploration vs Exploitation, Formulation of RL agents for few examples	T	CO3	B1 and Online

3	9-12	Finite Markov Decision Processes	Agent-Environment Interface, Goal and Rewards, Returns and Episodes, Task (Episodic and Continuous), Policies Value Function and Examples. Multi-arm Bandit Introduction	T	CO3	B1
4	13-14	Open AI Gym	Working understanding on Open AI Gym	L	CO3	Online
Assignment 1 – RL Application, Environment Exploration, State Space Design , RL Formulation						
5	15-19	Dynamic Programming	Introduction, Policy Evaluation, Policy Improvement, Policy Iteration, and Value Iteration	T/L	CO2, CO4	B1
5	20-21	Course Project Problem Identification	Choosing the application area and open problem in RL state-of-art	L	CO3	Online
6	22-25	Monte Carlo Methods	Monte Carlo Prediction, Monte Carlo Estimation of Action Values, Monte Carlo Control and Implementation	T/L	CO2, CO4	B1
7	26-31	Tic-Tac-Toe Implementation	Code Exploration and Implementation of Tic-Tac-Toe game and Evaluation	L	CO2, CO4	Online
8	32	Temporal-Difference (TD) Learning	TD Prediction, Advantages of TD and Optimality of TD (0)	T	CO2, CO4	B1
8	33	TD-Sarsa	On-Policy TD Control	T	CO2, CO4	B1
	Assignment 2: RL Algorithm Exploration DQN					
9	34-39	Presentation	RL Formulation and Algorithm - Use case	L	CO2, CO3	Online
9	40	TD-Q-Learning	Off-Policy TD Control	T	CO2, CO4	B1
Quiz-1						
Mid Semester Exam						
10	41-42	Q-Learning Tic-Tac-Toe Implementation	Implementation and Evaluation of Tic-Tac-Toe game using Q-Learning	L	CO2, CO4	Online
10	43	TD-Sarsa	Expected Sarsa	T	CO2, CO4	B1
11	44-46	n-step Bootstraping	n-step TD Predication, n-step Sarsa, n-step Off-policy Learning	T	CO2, CO4	B1
	Lab Evaluation					

12	47-50	Tabulator Methods	Planning and Learning using Tabular Methods	T/L	CO2, CO3, CO4	B1 and Online
			Quiz-2			
13	51-54	Deep Reinforcement Learning	Design of NFQ	T/L	CO2, CO3, CO4	B1
14	55-58	Architectures	DQN, DDQN	T/L	CO2, CO3, CO4	B1
15	59-62	RLHF	Usecase	T/L	CO2, CO3, CO4	B1
		End Sem	**Term Project**		**CO1, CO2, CO3, CO4**	

***Bi - Text Book / Reference Books i**

***Online Resources links in the Lab Assignment Details**

Text Books / Reference Books

1. 'Reinforcement Learning', Richard.S.Sutton and Andrew G.Barto, Second edition, MIT Press, 2018
2. Csaba Szepesvári, Algorithms for Reinforcement Learning, Morgan & Claypool. 2010.
3. Marco Tiering and Martijn van Otterlo, Reinforcement Learning: State-of-the-Art Adaptation, Learning, and Optimization, Springer, 2012.

Evaluation Pattern:

Internal: 50%

Component Name	Weightage/event in %	Maximum Marks	No. Of Events	Total Weightage in %	Exam Week
Quiz 1	5	20	1	5	1st week of September'26
Quiz 2	5	20	1	5	1st week of October'26
Mid Semester	20	50	1	20	1nd week of September '26
Assignment 1	5	20	1	5	4th week of July'26
Assignment 2	5	20	1	5	4th week of August '26
Lab Evaluation	10	20	1	10	1st week of October '26
Total				50	

External (Hybrid): 50%

Component Name	Weightage/event in %	Maximum Marks	No. Of Events	Total Weightage in %
Course Project	30	30	1	30
Written	20	20	1	20

Course Assessment Plan (CAP)

Course Outcome	Continuous Assessment Weightage						End Semester Weightage (Project)	End Semester Weightage (Written)	CO wise Total Weightage
	Quiz1	Quiz2	Mid Sem	Assignment 1	Assignment 2	Lab Evaluation			
CO1	2.5	0	4	2.5	2.5	0	5	4	20.5
CO2	0	5	8	0	2.5	0	5	6	26.5
CO3	2.5	0	3	2.5	0	0	09	6	23
CO4	0	0	5	0	0	10	11	4	30
Component wise Total Weightage	5	5	20	5	5	10	30	20	100

Course Project:

Evaluation Criteria

Parameter	Marks
Design (CO2, CO3)	10
Implementation, Results and analysis (CO3, CO4)	10
Viva (CO1, CO2,CO4)	10
Total	**30**

Evaluation Parameter	Assessment Criteria	CO Mapping	Marks
Design		**CO2, CO3**	**10**
Problem Selection and Definition	Identification of a relevant Reinforcement Learning problem and its practical significance.	CO3	2
Problem Formulation	Formulation of the problem as a Reinforcement Learning problem by identifying the agent, environment, state space, action space, reward function, and objective.	CO3	4
Algorithm Selection	Selection and justification of an appropriate Reinforcement Learning algorithm for solving the formulated problem.	CO2	3
Solution Design	Overall design and workflow of the proposed RL solution.	CO2	1
Implementation, Results and Analysis		**(CO3, CO4)**	**10**
Implementation	Implementation of the formulated	CO4	4

	Reinforcement Learning solution.		
Experimental Results	Successful execution and presentation of results obtained from the implemented RL solution.	CO4	3
Analysis and Interpretation	Analysis of the obtained results and discussion on the effectiveness of the proposed solution.	CO3	2
Conclusion and recommendation	Conclusion and Future scope	CO3	1
Viva Voce		**CO1, CO4**	**10**
Understanding of Reinforcement Learning Concepts	Demonstrates understanding of Reinforcement Learning concepts and explains how RL complements other Machine Learning techniques.	CO1	3
Understanding of Reinforcement Learning Algorithms	Understanding of various RL algorithms	CO2	1
Understanding of the Implemented Solution	Explains the implementation methodology, chosen algorithm, and obtained results.	CO4	4
Technical Discussion	technical communication, and ability to answer questions.	CO1	2

Lab Assessment Assignments Details

Lab Assignment 1
Title: Explore OpenAI Gymnasium

Activities:

- Install Gymnasium.
- Explore Tic-Tac-Toe 3D Environment
- Understand state space, action space, reward structure and termination conditions.

Students are encouraged to explore any two of the following environments

1. CartPole: https://gymnasium.farama.org/environments/classic_control/cart_pole/
2. Taxi: https://gymnasium.farama.org/environments/toy_text/taxi/
3. Blackjack: https://gymnasium.farama.org/environments/toy_text/blackjack/
4. MountainCar: https://gymnasium.farama.org/environments/classic_control/mountain_car/

Lab Assignment 2
Title: Reinforcement Learning using Tic-Tac-Toe

Activities:

- Explore the RL implementation.
- Observe agent learning.
- Modify parameters and compare results.

Reference: https://jinglescode.github.io/reinforcement-learning-tic-tac-toe/

Lab Assignment 3
Title: GridWorld

Activities:

- Implement policy evaluation.

- Implement value iteration.
- Compare shortest paths.

Reference:

- https://deeplearning.neuromatch.io/tutorials/W3D4_BasicReinforcementLearning/student/W3D4_Tutorial1.html
- https://github.com/ShangtongZhang/reinforcement-learning-an-introduction

Lab Assignment 4
Title: MATLAB Reinforcement Learning Onramp
Activities:

- Complete the MATLAB Reinforcement Learning Onramp.
- Train an RL agent.
- Analyze learning curves

Lab Assignment 5
Title: Reinforcement Learning Application Exploration and Performance Analysis
Activities:
Task 1: Select an RL Application (Autonomous Driving, CartPole Balancing, Taxi Route Optimization)
Task 2: Problem Formulation (Identify and describe: Agent, environment, State space, Action, Reward)
Task 3: Algorithm Exploration (Q-Learning, SARSA, Deep Q-Network (DQN))
Task 4: Execute the Application (Key: Training, reward obtained, number of episodes, success rate)
Task 5: Performance Analysis (Analyze: Learning curve (Reward vs Episodes), Agent performance)
Task 6: Demonstration (Present the application and explain: Working principle, RL formulation, Algorithm used, Experimental observations, Conclusions)

1. Learn the fundamentals of machine learning and reinforcement learning in a fun and engaging way through autonomous driving with AWS DeepRacer
   https://www.udacity.com/course/aws-deepracer--ud014

2. https://github.com/mimoralea/gdrl/blob/master/notebooks/chapter_02/chapter-02.ipynb

3. https://github.com/mimoralea/gdrl/blob/master/notebooks/chapter_03/chapter-03.ipynb

4. https://github.com/mimoralea/gdrl/blob/master/notebooks/chapter_04/chapter-04.ipynb

5. https://github.com/mimoralea/gdrl/blob/master/notebooks/chapter_05/chapter-05.ipynb

6. https://github.com/mimoralea/gdrl/blob/master/notebooks/chapter_09/chapter-09.ipynb